PCA + Linear MPC

01 Offline model fitting

Frozen Encoder

Training data (o_t, u_t, o_{t+1})

$z_t = E(o_t)$: 512-D latent

Fit PCA (train only)

Compare ranks: 16, 32, 64

Fit P, center μ, scale D

$x_t = D^{-1}P^T(z_t - \mu)$

Fit Ridge Regression

Inputs: x_t, u_t

Target: x_{t+1}

Selected Model: 64-D

$$x_{t+1} = Ax_t + Bu_t + c$$

$$A : 64 \times 64 \qquad B : 64 \times 2$$

15 rank / ridge candidates → Selected rank: 64

Ranks: 16 / 32 / 64; ridge: 1e-6 / 1e-4 / 0.01 / 0.1 / 1

Select by validation multi-step latent rollout MSE

02 Linear MPC

Encode & Project

Frozen Encoder + selected PCA

512-D latent → 64-D state

Predict with A, B, c

Candidate actions $u_{t:t+H-1}$

Roll out $x_{t+1:t+H}$

Optimize Actions

Track goal: $\hat{z} = \mu + PDx$

Action penalty + norm bounds

Execute First Action

Observe environment response

Update state and replan

Observation feedback · Execute one step, then replan